

Machine Learning Assignment - 1

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Datasets: var1 (Steam Turbine, 6 inputs), var2 (Thermal Reservoir, 3 inputs)

GitHub: <https://github.com/anon-k/Machine-learning-Assignment-1>

Approach (both phases)

1.1 Polynomial regression as linear regression:

Inputs are expanded into all terms of total degree $\leq d$, and a linear model is fitted on those terms. The number of terms is $C(d+n, n) - 1$, which grows quickly. Phase 1 has 461 terms at $d=5$ and 8,007 at $d=10$. Phase 2 has 164 terms at $d=8$ and 1,770 at $d=20$. With only 1,000 training rows, high degrees have more unknowns than data.

1.2 Preprocessing

- Phase 1: standardise the inputs, expand to polynomial terms, then standardise the expanded terms so the regularisation penalty treats every term fairly.
- Phase 2: scale the inputs to $[-1, 1]$ and use a **Legendre polynomial basis** instead of raw powers. Legendre polynomials are nearly uncorrelated on $[-1, 1]$, which avoids the numerical instability of high-powers. They span the same set of polynomials, so nothing is lost.
- Scaling uses input features only, never y .

1.3 Regularization - Two penalised models were compared:

- **Ridge (L2):** shrinks all coefficients. The penalty strength α is chosen by built-in cross-validation (**RidgeCV**).
- **Lasso (L1):** shrinks coefficients and sets useless ones to exactly zero, giving automatic term selection. α is chosen by 5-fold CV (**LassoCV**)

1.4 Degree selection

- Every (method, degree) pair was scored with 5-fold cross-validation (shuffled, seed 42, same folds for all candidates). I recorded training and validation MSE and R^2 . The pair with the **lowest validation (CV) MSE** was chosen. $R^2 = 1 - \text{MSE}/\text{Var}(y)$, so the two metrics rank models identically, and MSE is used for selection while both are reported.
- Training error is never used for selection, since it always favours the highest degree.
- Lasso was run only on degrees near the Ridge optimum because it becomes very slow with thousands of terms.

1.5 Sanity check

The chosen model was trained on 80% of the data and scored on the remaining 20% (hold-out). It was then refit on all 1,000 rows to predict the test set.

Phase 1: Power Plant Steam Turbine (var1)

2.1 MSE and R^2 values for both Ridge and Lasso regularization, across degrees 1-10

Degree	Ridge MSE	Ridge R^2	Lasso MSE	Lasso R^2
1	9.01942	0.08815	-	-
2	3.2765	0.66645	-	-
3	1.09904	0.88735	1.07106	0.89037

4	0.79305	0.91823	0.67983	0.93027
5	0.53475	0.94477	0.34370	0.96474
6	0.60948	0.93689	0.36503	0.96263
7	0.61689	0.93646	-	-
8	0.72954	0.92458	-	-
9	0.77490	0.92030	-	-
10	0.90077	0.90697	-	-

(– means not run. Lasso is slow at high degree and Ridge was already worse there)

2.2 Plots

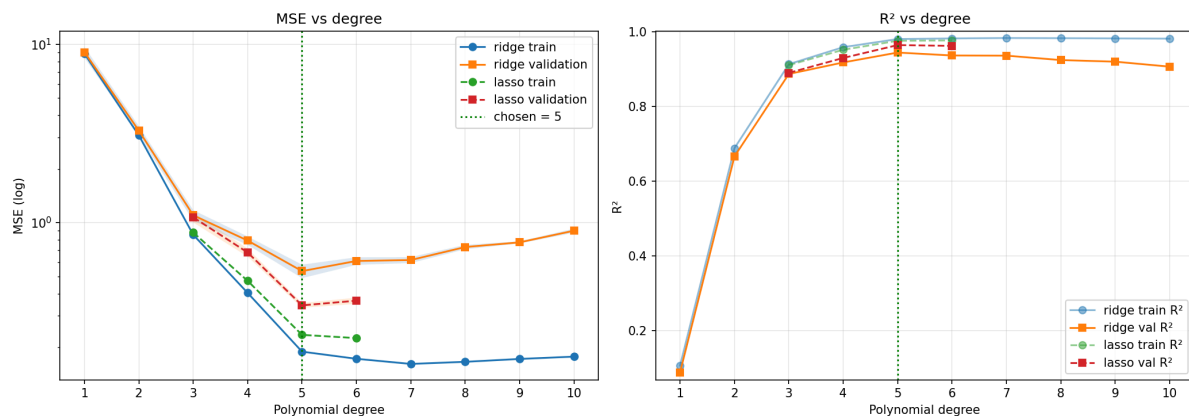


Figure 1. Left: training and validation MSE vs degree (log scale). Right: training and validation R^2 vs degree. Ridge is drawn solid and Lasso dashed. Circles are training, squares are validation. The shaded band is ± 1 standard error of validation MSE across the 5 folds. The green dotted line marks the chosen degree (5).

Left panel (MSE):

- **Training curves (circles)** fall with degree and then flatten. More terms always fit the training data better, so this curve cannot be used to choose the degree.
- **Validation curves (squares)** are the ones that matter. They fall steeply from degree 1 to 5, reach their lowest point at degree 5, and then stop improving. Ridge validation rises again from 0.53 to 0.90 by degree 10. Lasso validation moves from 0.344 to 0.365 at degree 6.
- **Regions:**
 - *Degrees 1 to 4:* training and validation errors are both high and close together. This is underfitting.
 - *Degree 5:* validation error is at its minimum.
 - *Degrees 6 to 10:* training error stays low while validation error climbs. The gap between them widens, which is overfitting.

Right panel (R^2): this is the same information flipped, because $R^2 = 1 - \text{MSE}/\text{Var}(y)$. Validation R^2 rises quickly, peaks at degree 5 (Lasso 0.965), and then flattens or slips (Ridge falls to 0.907). Training R^2 keeps creeping up toward 1, again showing that training score alone is misleading.

2.3 Chosen model: Lasso, degree 5

CV MSE	CV R^2	Hold-out MSE	Hold-out R^2
0.34370	0.96474	0.32183	0.96586

2.4 Rationale

- Degree 1 has $\text{MSE} \approx 9.0$, about the variance of y , so the relationship is almost entirely nonlinear.

- Error falls sharply up to degree 5, with the largest drop from degree 4 (0.68) to 5 (0.34), which indicates genuine degree-5 structure.
- Beyond degree 5, error does not improve (Lasso degree 6: 0.365) and Ridge gets worse, so degree 5 is the simplest model at the minimum.
- Lasso beat Ridge. Of the 461 degree-5 terms, about 129 had nonzero coefficients.
- Hold-out and CV agree, so there is no sign of overfitting.

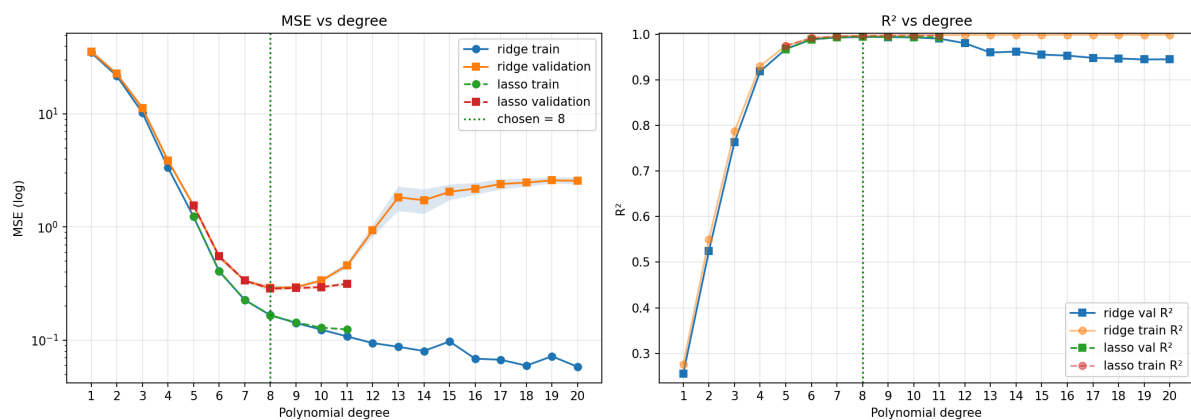
3. Phase 2: Subterranean Thermal Reservoir (var2)

3.1 MSE and R^2 values for both Ridge and Lasso regularization, tabulated for selected degrees in 1-20

Degree	Terms	Method	MSE	R^2
1	3	Ridge	35.42	0.256
4	34	Ridge	3.87	0.918
6	83	Ridge	0.548	0.988
7	119	Ridge	0.335	0.993
8	164	Ridge	0.287	0.9939
8	164	Lasso	0.285	0.9939
9	219	Lasso	0.287	0.9939
10	285	Lasso	0.294	0.9937
12	454	Ridge	0.928	0.980
15	815	Ridge	2.040	0.955
20	1,770	Ridge	2.552	0.945

(Ridge was run for degrees 1 to 20. Lasso was run only on degrees 5 to 11, around the Ridge optimum)

3.2 Plots



Left panel (MSE):

- **Degrees 1 to 7 (underfitting):** training and validation curves lie almost on top of each other and fall steeply (validation MSE 35.4 to 0.33). The model is too simple, and the error is not coming from noise or memorisation.
- **Degrees 8 to 10 (flat minimum):** validation MSE is at its lowest and nearly flat (0.285 to 0.294). The differences are smaller than the shaded band, so these degrees are statistically tied. The simplest of them, degree 8, is chosen, and it is also the numerical minimum.

- **Degrees 11 to 20 (overfitting):** the training curve keeps falling (0.108 to 0.058), but the Ridge validation curve shoots up (0.46 to 2.55). This widening gap is the overfitting signature. With 1,770 terms and only 1,000 rows, the model has more parameters than data points.
- **Training error never reaches zero**, even with more terms than rows. This suggests random noise in y . The validation floor of about 0.28 gives a rough noise-level estimate. This is inferred from the curves, not given.

Right panel (R^2): validation R^2 climbs from 0.26 to a plateau of about 0.994 at degrees 8 to 10, then falls to 0.945 as overfitting sets in. Training R^2 stays near 0.998 throughout. The right panel shows how little there is left to gain after degree 8, because R^2 is already about 0.994. It cannot separate degrees 8, 9 and 10, whereas the log-scale MSE panel can.

3.3 Chosen model: Lasso, degree 8

CV MSE	CV R^2	Hold-out MSE	Hold-out R^2
0.285354	0.993906	0.356421	0.990900

3.4 Rationale

- Train and validation errors are both high and close together below degree 8 (underfit), and diverge sharply above degree 11 (overfit).
- Degrees 8 to 10 are statistically tied, so the simplest was chosen.
- The reported maximum degree of 20 is therefore not estimable from 1,000 noisy points, and degree 8 is the best *estimable* model.

4. Comparison and discussion

- Neither dataset needed anywhere near the maximum allowed degree: the best degrees are 5 (of ≤ 10) and 8 (of ≤ 20).
- Regularisation was essential, because the number of terms approaches or exceeds the number of rows at higher degrees.
- Selecting on training error would always pick the largest degree. Using cross-validation error is what prevents overfitting.
- **Limitations:** CV estimates carry noise (differences below about 0.01 are not meaningful), and Lasso at high degrees was not explored because of runtime.

5. Final results summary

Problem	Model	Degree	CV MSE	CV R^2	Hold-out MSE	Hold-out R^2
var1	Lasso	5	0.34370	0.96474	0.32183	0.96586
var2	Lasso	8	0.285	0.994	0.356	0.991

6. Additional information

- Libraries: NumPy, pandas, scikit-learn, Matplotlib.
- Fixed seeds: `KFold(random_state=42)`, hold-out `random_state=0`.
- Files: `phase1.py`, `phase2.py`, the two CSVs `phase1_cv_results.csv` and `phase2_cv_results.csv`, the two prediction CSVs, and the figures.